

ADR-001: Architectural Style Decision

Status

Accepted

Context

The project aims to build a real, production-deployable restaurant operations system that may evolve into a SaaS platform.

The system must:

- Be maintainable by a single developer
- Support clean separation of responsibilities
- Allow future scalability
- Be deployable on AWS
- Demonstrate architectural maturity for portfolio purposes

Possible architectural options considered:

1. Traditional layered monolith
2. Microservices from the start
3. Modular monolith with Clean Architecture

Decision

We will implement a:

Modular Monolith using Clean Architecture principles.

Rationale

Why NOT microservices?

- Operational overhead (networking, service discovery, observability)
- Increased deployment complexity
- Overengineering for MVP
- Not aligned with single-developer constraints

Why NOT traditional layered monolith?

- High coupling risk
- Controllers directly depending on persistence
- Poor testability
- Violates Dependency Inversion Principle

Why Modular Monolith + Clean Architecture?

- Clear separation between Domain and Infrastructure
 - Use cases are independent of frameworks
 - Easier to test
 - Easier future extraction into microservices
 - Aligns with SOLID principles
 - Production-ready structure
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Architectural Layers

Domain

- Entities
- Value Objects
- Business rules

Application

- Use Cases
- Interfaces (Repository contracts)

Infrastructure

- Prisma repositories
- Database access
- External adapters (Siigo, WhatsApp)

Interface

- HTTP controllers
 - Guards
 - DTO validation
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Consequences

Positive

- High maintainability
- Clear module boundaries
- SaaS-ready evolution
- Good portfolio demonstration
- Testable business logic

Negative

- Slightly more boilerplate than simple monolith
 - Requires discipline to avoid layer leakage
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Future Considerations

If the system evolves into SaaS:

- Introduce tenant isolation
- Add billing service
- Extract integrations into independent services if needed

Decision Owner

Project Author